

ActInf GuestStream 078.1 ~ Juan Diego Bogotá:

"What could come before time?"

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hello and welcome it's April 3rd 2024 we're here in active inference guest stream 78.1 discussing what could come before time with Juan Diego Bogata and this should be a great presentation followed by a discussion so thank you again for joining looking forward to the presentation and discussion thank you Daniel so uh hello everybody so my name is ju and today I'm going to talk about mainly phenomenology no so much about not so much about active inference but maybe all discussion will go that way I kind of assume so to give you a bit of context about this so uh this presentation is actually based on a paper that I recently published in phenomenology and the Cog Sciences uh last month but to give you a bit more context uh in general I'm interested in the relationship between cognitive science and phenomenology and when I talk about phenomenology I'm very mindful about the fact that this is a term that is a bit ambiguous in the literature because if you talk to maybe philosophers of mind or cognitive scientists when they talk about phenomenology they often think about say qualia or you know the what it's like of uh phenomenal Consciousness and that's okay but I usually use that term to refer to specific philosophical tradition and known as phenomenology that was started uh towards the beginning of the 20th century by Edmund husel and it was further developed by people like meron or haiger or Simon De and the whole point of phenomenology is to try to provide a very strict and rigorous analysis of Consciousness and live the experience uh by taking very very seriously the first person perspective so it's a matter of trying to unveil the invariant structures of Consciousness and I think that this methodology is quite promising for us to uh better understand uh cons Consciousness and cognition so it might be an interesting thing to further integrate into our scientific understanding of the mind and I think that a good way of connecting phenomenology with the broader kind of iral context of the of the cognitive Sciences uh is the inactive approach mainly because uh people like Francisco Aela or Evan Thompson were very have been very influenced by phenomenologists like whoer andan when it comes to ideas like

embodiment uh the embeddedness of cognition um the importance of subjectivity these kind of stuff and in this particular talk I want to focus on uh something that these two Frameworks have in common which is uh how they characterize intentionality so usually uh in cognitive science philosophy of mind when we talk about intentionality we mean kind of this directedness or aboutness of certain mental States so there's this kind of uh relationship between the mind and specific object in the world and that's that's uh kind of um uh the usual concept of intentionality but I think that phenomenologists and quite a bit of activists have a bit of a broader conceptualization of intentionality not only as this aboutness that is directed towards a specific object in the world but rather as a general openness to the world and the whole idea of phenomenology is precisely to identify the structures that uh constitute that openness and that's again that's kind of the context of my presentation and my paper uh but when you look at the kind of the literature in classical phenomenology and you notice that there's one particular structure or feature of lived experience that has some sort of Primacy and that's temporality or as whoer calls it time Consciousness so here I bring a quote of P what do you mean there's like multiple kind of overlaid oh hold on visual frames I'll try again thank you I was entertaining the possibility it was intentional oh no yeah I don't know what happened let me try again yeah looks good thank you okay fantastic okay so now without the overlay so here have H quote of feral talking about time Consciousness uh he says that in the ABCs of the Constitution of all objectivity given to Consciousness and subjectivity as existing for itself time Consciousness is the a if consist as we might say in a universal formal framework in a synthetically constituted form in which all other possible syntheses most participate and this a very kind of fancy and worthy way of saying that you know if you want to understand Consciousness uh from its first person perspective the first thing you have to analyze is temporality that's kind of the basis of it all and the reason why uh we should start by temporality is because not only we tend to experience things as unfolding in time but also because uh Consciousness itself is a stream of Consciousness it's a temporal unfolding and that kind of uh frames all of our experiences so temporality is kind of the the fundamental structure of of Consciousness uh and when we get deeper into the hustan analysis of temporality we tend to find this uh triarte structure of time Consciousness so who tend to say that what he calls the Liv the living present which is literally how experience the present right now is usually an integration of the past the present and the future the just past the immediate present and what is about to come and these have like very specific terminologies we call a retention the kind of this experience of the just past paral impression experience of what is

literally happening right now in like a very uh small uh now phase of perception and protention to the anticipation of what it's about to come and well these different kind of operation of Consciousness have very complex uh relationships I'm not going to dwell too much into them the whole point is that they together constitute the kind of the flowing feature of Consciousness how it flows temporally how it's kind of um sinking into the past because we on we're constantly anticipating what is about to come this anticipation is usually implicit we don't necessarily uh think about what is about to come explicitly it just kind of happens but when you experience something this uh like the the Primal impression either confirms or fulfills what we were pretending or anticipating or negates the anticipation and regardless of that whether there is anticipation or fulfillment uh what is experienced kind of sinks into the past and this is called retentional modification so we have this kind of continuous flow of time that goes from the future towards the past sinking into the past but importantly this um this constitution of of the living present or the flow of time can be be conceptually understood as pretemporal or atemporal because the whole point is that this these processes these operations are literally constituting our experience of time so they cannot be given in time such because if you if you claim that then you have um uh a circular argument uh who nevertheless recognizes that this structure the structure of time Consciousness is just an abstraction it's just a formal framework that has to be further um complemented by qualitative content so even though time time Consciousness kind of the basis of it all we have to go beyond in order to better understand Consciousness and when we talk about content we have to talk about affection so for feral affection is the lure given to Consciousness The Peculiar pool that an object given to Consciousness exercises on the ego it is a pool that is relaxed when the ego turns towards it attentively so it's just like in in very simple terms affection occurs when something in our field of of perception our perceptual field kind of tries to catch our attention we feel this kind of solicitation or pull or tendency or disposition to pay attention to something um and for Co this this process of affection corresponds to the basic form of intentionality because that's literally when we first have this directness towards an object it's a first moment which is an object becomes Salient for us and it also gives rise to the minimum form of activity which is attention so there's this distinction between affection and attention for me to attend something to being able to uh being attentive towards an object that object should try to call my attention has to affect me beforehand logically not temporal it's like a logical relationship in here so that's why we can H conceive of affection as this kind of precondition of attention and to better give you uh an illustration of what affection is you can see this it's the the flag of Japan

and probably when you see the flag of Japan the thing that catches your attention the most is the red circle in the middle and the reason why this catches our attention is because of the very Stark um contrast between the circle and the background the white background so in a lot of cases very like in sensorial cases affection is related to this contrast compare this to saying um an artwork by Jackson Pollock in this case there's there's a bunch of pain strokes and there's nothing that is specifically catching your attention maybe you're seeing the yellow then the red and the black it's like chaos and and the reason why this happens at least from phenomenological perspective is because uh there's no there's no there is contrast but it's not as strong as in this case of course again this is like a very basic case we can also introduce uh interest and concerns sometimes something catches our attention just because uh it's relevant for our current interests so for instance now I'm very focused on the screen of my laptop and it's precis because I'm interested in giving this presentation so this is broadly what affection means this kind of felt felt P or tendency towards an object or like a sensory content in our con in our uh perceptual field now the importance of affection lies in the fact that the world is first properly given and experienc through it for something to be experienced it must affect me so this kind of shows the importance of of affection in relation to temporality temporality provides the form framework that organizes uh experiential content but for us to experience something at all there has to be affection and I like to to link the the hustan concept of affection to the broader con concept of affectivity and I borrow jaavan colet's concept of affectivity it's not only about emotions or mood or this different kind of effective States but rather it's a broader Dimension that she defines as a lack of indifference and in this case we can say that affection in theeran sess is completely consistent with affectivity because when something affects us then we are no longer indifferent to it so that's effectivity in the broadest and most basic of Senses but importantly uh for H and I would say that for other classical phenomenologists affection still presupposes time conscience temporality mainly because uh for something to affect us uh it has to occur has to be given in the impressional present so then our phas it also affirms this orientation toward the future this tendency towards attention it's solicitation towards the future so it presupposes protention and there's also a relationship with retention but I'm not going to dwell too much into that the whole point about this is that uh given that affection still presupposes time Consciousness we can say this H would say temporality is the most basic uh structure of uh Consciousness so if we can want to do cognitive science or Consciousness research we should probably start by temporality and interestingly if we go to uh the an active approach and velas neurop phenomenology which is

a way of trying to integrate phenomenological research with uh neurobiological research by establishing reciprocal constraints between these two domains uh the first neurophenomenological study and Analysis that barela developed is precisely of time Consciousness and he considers it to be kind of an acid test for the neurophenomenological project but there's some differences between how barela and Hil conceive of temporality whereas for Hil temporality is kind of abstract formal framework that is always organizing our experiences Vela very clearly states that embodied activity gives temporality its roots in living itself so that's kind of an active gimmick isn't it uh we have for Vela what actually gives rise to experience of time is the fact that we're embodied beings that are um relating their where interacted with the world through action and an example of this is something like the do grait illusion so sometimes you see a dog sometimes you see a rabbit and but I would say that this shift which is a temporal shift we can actually experience kind of the temporal unfolding of how our percept uh changes over time for rather lot this shift uh occurs because there's action going on usually um very minimal and often unnoticed action for for example visual Sackets or you know maybe just Shi your attention a bit and that changes the whole percept and this kind of shows for bellaa that we actually have to to focus on this embodied activity that gives rise to this constant temporal enfold of how we perceive the world and the kind of the whole point of neurophenomenology is to try to look at what's happening at the neural level and what Vela says or he he discovers is that if you look at the neural dynamics that uh underpin this embodied activity and you describe them through the tools of dynamical System Theory the kind of model you get you know uh trajectory in state space these Dynamics would mirror will probably mirror the structure of the living present as it is analyzed by fer so we have this kind of almost isomorphic relationship between the Dynamics of uh the neurodynamics that underpin embod activity and the Dynamics of time Consciousness which for barela suggests that that's precisely where we can find the reciprocal constraining between phenomenology and neurobiology uh but there is an important distinction between another important distinction between Vela and hell here whereas for hell at least in the classical texts in which he analyzes time Consciousness protention is kind of the mirror opposite of of retention so pretention is kind of this retention that it's looking to the Future buta notices that there's something different about protention for him protention uh seems to be marked by an affective tonality and in in the in the 1999 paper the example he gives is that what he's trying to write this work with what is trying to find the correct words he feels this this tendency this disposition to look for a very specific word but it's also marked by frustration because he

cannot really find the proper uh expression so there's this kind of correlation between anticipation and the tendency towards the future and a feeling tonality and that's all he says in the N 199 1999 paper he says that I'm just doing a sketch of this hopefully I'll work on this in the future and I think there's two ways in which we can find kind of the elaborations on these ideas on the one hand we can borrow from uh later and active analysis of emotion uh when you read for example the work of Evan Thompson or Javanna Coletti they often emphasize the fact that emotion uh is a process that integrates a lot of neural and processes including attention evaluation bodily arousal and importantly for what I'm interested in Act Tendencies and a way to to uh illustrate this is by borrowing Walter Pan's Dynamic architecture of the lyic system so for Freeman if you look at the Dynamics of brain body and environment you can find a lot of causal Loops that constitute H certain cognitive processes so for instance if we want to understand action we have to understand how sensory systems the hippocampus the motor system systems and the environment from this causal loop it's an action perception Loop mainly and this suggests like that environment body and brain co-constitute uh action cognitive action but the important thing here for me is that all these Loops all these different uh proceptive M Loops all go through the hippocampus which is as you probably know kind of the one of the most important parts in the brain that is related to emotion and this is a way of noticing that all these uh cognitive processes uh action perception proprioception are probably at the sub personal neural level uh colored or constituted by emotion and what H Thompson takes from this is that if you look at the relationship between the motor system and the sensory system there has to be uh what we may call a state of expectancy because our sensory systems have to be prepared uh in relation to the consequences of our actions and this is kind of very very similar to some of the claims in the active inference literature although this is earlier but that's important because then you have have this kind of anticipation what it's about to come at the sensory level at least at a neural level and Thompson says that we can actually link this neural process to what at the experiential level phenomenologist called protention so we start seeing this connection between protention so temporality and emotion so affectivity so we see at a neurophenomenological level how we're moving beyond just this very formal and abstract description of temporality but even if we look at this paper by Banda in 2005 they go further and they analyze the experiential level of affection and they kind of analyze it in in a form as a form of a what they call a microtemporality so they say that if you analyze not only kind of the neural level but also the phenomenological SL experiential level you'll find that there's

this kind of very quick uh integration of events you have the the precipitation of an event or trigger so something affects you uh the emergence of salience so experientially what affects you become Salient and there's a balanced feeling tone that's the call effective tality I've been talking about for a while already there's motor changes so there's action tendencies that are formed towards the object and there's autonomic physiological changes that kind of bring everything together and this kind of this micro temporality actually constitutes what US calls affection but from this analysis which is completely consistent with the neurop phenomenol analysis I just mentioned uh V Anda claimed that uh this micro temporality implies that affection precedes temporality because it gives rise to a tendency that can only deploy itself in time and thus as time and that's a very cure sentence for me to be honest I had to read it like a thousand times when I was reading that paper um but from what what I can understand and I think that um at least Evan Thompson seems to read it very similarly uh is that what happens with this micr Pro is that it terms a tendency an action tendency that precisely is protention so what we see is that the the this integration of very quick events at the that is correlated to the neurophysiological level are actually what give rise to to protention and because protention is kind of the spearhead of temporality we can say that affection precedes temporality is kind of the temporal and logical uh and empirical um antecedent to temporality so we have a very different claim uh in comparison to what Hil and other other phenomenologists say whereas Hol would say you know the first the most important and most basic uh structure of Consciousness is temporality it seems like some activist would say no no no no temporality is super important but it actually presupposes affectivity and I think there's something odd about that claim because how can affection preced temporality if it is deployed in time there seems some sort of paradox going on here how can that's precisely their claim if you see here they say affection temporal because it gives R to to a tendency that can only deploy itself in time and thus as time that's they're claiming temporality emerges in this micro temporality of affection and I think that there's the risk of um conflicting two different temporalities in here there is on the one hand what we may call objective time so it's the time of neural processes the time that neuroscientists use to measure all these um neural events and Dynamics and it subjective time so time Consciousness how we experience the world and it's problematic to conflate these two kind of temporalities because the whole point of phenomenology from the cons perspective is to find the structures that actually allow us to experience time as an objective um feature of the world and when I say that there's the risk of completing t two levels in the inactive uh approach it's very clear when you see this precisely the microt

temporality if you take a look at this microtemporality some events are clearly experiential for instance the emergence of sense and the valence feeling tone so we experience the salience of objects we experience the feeling tonality of what it's like to be you know a human being that engages with objects but for instance motor changes and autonomic physiological changes seem to be more of a subpersonal or even neural process so that's at the not at the subjective level but rather at the objective level so there's this kind of shift in between the objective and subjective levels which I think is what actually gives kind of justifies the claim that affection precedes temporality I mean affection prior to temporality if we understand affection just as at this level at the level of neural process and neural loops but if we look at it from an experiential perspective that is not so clear because we cannot really experience these loops we experience kind of the effect you want to put it like that or kind of what emerges from these correlations we don't experience them as such so again saying that affection prior to temporality it is as if we are saying affection a subjective process objectively precedes another subjective process and that's where the completion kind of uh comes in and what is interesting about this this completion is that if you look at the 1999 paper of Vela the one on time consciousness he realized there's something odd about the relationship between temporality and affectivity or emotion as he calls it there and he says that maybe what we have to look for is a what he calls a nonlocal solution or a non-dual perspective so we're trying to show that these two things are not two things there maybe one or there's some sort of dialectic going on if you want to put it like that and it's quite interesting because again at that point he's saying I'm just sketching some ideas and then later on when he publishes the Vela and the paper of 2005 which is actually like 2002 um there's no non-dual perspective at all what you find is precisely a dual perspective According to which affectivity precedes temporality so what I want to do here now is to try to pursue this perspective and show that temporality and affectivity go hand in hand and they're co-emergent but to actually show that we have to avoid conflating the objective and the subjective levels and that's the reason why I focus on the phenomenal level so this brings me back to uh H so when we're talking about affection it's very important to notice that uh this kind of transition from affection to attention gives rise to uh foreground background structure in perception and it's very well noted by Gestalt psychologists later on so for instance in this usual example of Gestalt perception if you focus on the black shape you will probably see something like a vase but the this perceptual meaning the fact that you're perceiving a black vase um it's only possible because there is also a white background so you have this kind of perceptual interplay

between the foreground and the background whereas the black uh foreground the object of attention is given in a specific way the background is still given you're experiencing the the white part but it's not the focus of your attention and that's very important because that kind of shows that when we shift from affection to attention there's still an affective background kind of not only pulling trying to pull our attention but it's also kind of giving meaning to what we're pering so that kind of shows that the the object of attention always presupposes an effective background and this not only happens in like very simple cases like this but also if we look at a more complex uh painting uh there's a lot of different objects and colors and beautiful kind of uh sketches and uh that are trying to precisely pull our attention it's affection and if you focus for instance in the face of this girl there's a lot of colors that are still there and they they will probably produce a feeling of I don't know maybe you like the painting may you think it's beautiful or maybe you don't like that but this all comes in the interplay between foreground and background or between attention and affection and if we go kind of zoom out from here and go back to the concept of intentionality then a phenomenologist like me would say that this kind of shows that we can distinguish two concepts of intentionality that are always at play there is what we may call object intentionality which is the usual way which Phenom in which philosophers of mind and cognitive scientists talk about intentionalities this kind of directedness or aboutness of the Mind there to specific object in this case it could be say the face of the girl that's the intentional object phenomenologists would say that that's an active attentive an explicit for intentionality it is attentive because it's it presupposes attention it's explicit because I'm explicitly related to that I'm thematizing an object and it's active because in a sense I am the one who decides what you attend at least at a very minimal level there's a form of action but behind this there's always this effective background or effective Horizon which is also intentional because we are experiencing it it's given or pre-given who would say but it's easier to think about as given in experience but it's passive because we do not decide what affects us it just kind of happens it's something in our perceptual field it's effective because it it um occurs through affection and it's often implicit because we are not paying attention to the background precisely because we are attending the foreground the background is often implicit but this this um background is super super important because it shapes how we experience the intentional object if you want to go further um deeper into into the concept of practive intentionality you can think about the experience of your body so prop perceptive experience we often do not pay attention to how our body feels but it's always there in the background it's affecting us it's like a form of self affection some phenomenologist

would say and this all comes together precisely to uh give rise to the specific experiential meaning of the intentional object at the level of object intentionality so this kind of shows that affection is way more important than perhaps more complex than initially thought and to even kind of com uh make things a bit more complex we can also take into account the fact that affection is deeply embodied so to give you an an example elaborated by H he says that every perception that presents the object to me in this attentional orientation lives open the Practical transition to other appearances of the same object so this practical transition to other appearance is precisely the the transition from affection to attention I can always um thematize or attend what is affecting me but it as as who would note this involves practical uh engagements so he says that the possibilities of transition are practical possibilities there is does of freedom to run through run through the appearances in such a way that I move my eyes my head alter the posture of my body go around the object direct my regard to it and so on so this shows that attent affection is not only a like it's not an abstract pool not an abstract solicitation It's actually an embodied solicitation it's not only trying to uh object trying to catch our attention and and that's it they're trying to move us in a very literal sense and this is kind of very very familiar to an activist this is precisely what Vela was trying to point out when he was saying that uh embodied activity is at the root of temporality this is the embod activity he was thinking about it's already found in the works of hro and if we take very very seriously this practical or embodied nature of of um affection I think that we end up thinking about the world as something very similar to um climbing wall so if you ever been to a Climbing Center you might encounter something like this walls full of uh holes and they have different shapes they have different colors some of them are very very difficult to get a hold of some of them are quite easy there's there's a lot of stuff going on and all of them are trying are are quite literally what ecological psychologist would call affordances they afford different uh action possibilities and how they afford how strongly they afford to do that how strongly they solicit your action depends on thing depends on things like uh your skills how skillful you are uh the kind of body you have or even the interest you have so if you're trying to run through a um a specific curse in the in the um hang in the climbing wall some holes will be completely irrelevant uh in relation to the curse you're trying to to do so they are not as uh affording as they are as they might be and that's kind of how I understand the world from a phenomenological perspective this is what operative intentionality opens up the world becomes what ecological psychologist would call a field of relevant affordances so again it's just some affordances are more relevant because they're important for our concerns our needs

our skills or the kind of BU body we have that's mainly the point and what I like about affordances is that you can very easily move from this more ecological perspective to a phenomenological perspective by asking how AFF forant perception works again that's a classical phenomenological question how does this specific form of experience Works what are its structured and and as I've done elsewhere I've analyzed this afford perception as and interplay between a pull or solicitation coming from the object so the object is pulling or solic soliciting a specific action and a subjective projection of practical possibilities and this subjective projection is precisely uh future oriented projection us would say that affection has a unitary tendency toward the future intentionality is predominantly oriented toward the future so we might say that objects appear as affording actions as soliciting action that's that's affection but these actions can only be done in the near future it's an I can in the future so the projection is not only bodily but also temporal is a form of bodily anticipation and this kind of starts intertwining affection and temporality in a very very nice way I think from this perspective I believe operative intentionality involves the interplay between effective solicitations and future oriented or potential projection of practical possibility ities The Horizon that serves as the background of attention must be experientially understood as a set of bodily anti Tendencies so this shows in which sense for me affectivity and temporality are co-emergent on the one hand affection presupposes the potential projection of bodily possibilities so projection h potention of temporality shapes affection but on the other hand temporality or more specifically prot tension and anticipation presupposes the effective disclosure of the world if nothing were to affect me there would be no impressional present from which anticipations are formed so objects are pulling me right now and it's because right now there's this effective tendency that I actually have these projections and this kind of shows that affection and protention the spearhead of temporality presuppose one another which shows that they're co-emergent which is precisely what both uh an activist and phenomenologist can of Miss I think and this uh I believe that um has very interesting consequences and to give you um illustration of what I mean let's talk about very very briefly about depression so here I'm borrowing a figure from a paper by sanan and colleagues in which they are trying to sketch different fields of relevant affordances uh you can ignore the OCD one I just want to focus on the quote unquote normal one and the impression one to each of these bars represents how soliciting an affordance is affordance is in the environment so for us if we're not suffering from um severe depression some objects or some events in the world will solicit us more than others there's a stronger effective tendency there's stronger need or

pull coming from the object why again depends on our concerns our interests our bodily skills
Etc what happens in depression is kind of there's a flattening of the field of relevant
affordances people with depression struggle a lot with trying to wake up in the morning or
doing everyday tasks because their world is not from their perspective is not kind of soliciting
all these actions and this can be represented visually with this figure and I find it very very
striking when you link this with actual um descriptions of what it's like to be in a severely
depressed state so for instance law describes his own experience in depression as follows in
depression every tasks take on an as on an aspect of impossibility I see a task as far more
difficult than it actually is or I see myself as not being up to the task so again it's an aspect of
impossibility whereas in quote unquote normal uh perception there's this of this kind of tacid
or PR effective and bodily I can what I can do is always related to these affordances in
depression it's a complete impossibilities I cannot and this again this completely illustrates
from experiential perspective how the field of relevant importances for the depressed person is
almost patent but is also links with experience of time in depression so here I have another
quote this is from lot uh he says that in in his description of his experience in depression he
says I have absolutely no faith in fact in anything in a mdy way I see that depression manifests
itself as a crisis of faith not religious Faith but the almost born Instinct that things are fluid that
they unfold and change the new kinds of movement the sorry the new kinds of moment are
eventually possible that the future will arrive I am in a Time locked place where the moment I
am in with I am in will stretch on agonizingly forever and what what is kind of fascinating for
me about these three points together is that they very very clearly show how uh affection so
this solicitation and temporality go hand in hand on the one hand the world is not soliciting it's
not motivating actions for the subject for thepress subject and this is also experienced as a lack
of future future is no longer there it is as if if they are time locked in place and this precisely
shows what I'm trying to to show that affection this this uh affordance this the solicitation go
hand in hand with how we experience time especially as future oriented and this also shows
another point which I further develop in my my paper but I'm not going to dwell too much
here that this relationship between affectivity and temporality is also expressed as what I may
call General feelings so for instance here lot talks about a a lack of faith and that lack of faith
is quite literally I would say a feeling in relation to the whole world so at a the operative level
of intentionality the world is opened up as lacking not only motivation but there's no faith
towards it which is more of a feeling domain but again we can discuss this later on I just want
to show here how in uh depression we can see that both uh affect or affection and temporality

are uh Disturbed so to finish off how can affection receive temporality if it is uh deployed in time well it doesn't if we if we keep ourselves at the experiential level I would say that given this kind of phenomenological analysis now we can go and try to understand empirically and neurobiologically the the neural underpinnings of of Consciousness but we have to take into account this correlation this co-emergence between temporality subjective temporality and affectivity uh and to answer the question of the title of the paper nothing comes before time but it comes hand in hand with affection and that's it thanks thank you thanks awesome thank you sharing okay first though I hesitate to use such a time bound word Dean um what are your what are your thoughts well first of all thank you and I apologize for showing up late ju Diego the last time I saw you you were working on some things and they was really impressive and so I I was really excited about um hearing what you had to present today I have a quick question for you um in the active infant Community right now we are working on trying to put together something that addresses the ontology of being inactive inference what does that mean what's the definitional piece of that what is it mean to be an active infer and the the idea that you raised which is trying to make sense of that with temporality front of mind okay so when you put your slides up for example you had all the images you had the Duck Rabbit there's a certain stability even though the agent can pull interpretations from that but there's a there's a kind of consistency a constant to that whereas if I'm being an active infer if I'm con the only constant is that I'm consistently updating I'm always changing I'm in that flow I'm in that condition and it's the back and forth between the relative stabilities and the and the degree of change that a person goes through in order to keep up to keep up with their Niche and understand that they're enveloped but as you pointed out there's also these isomorphisms and this isovist perspective and this ability to maybe break timeout into more than one type so I was just wondering based on the work that you've done here if you were to try to come up with an ontology how would you be able to bring together the sense of consistency and the consist and the consistency of change that is a fascinating question and I and I think that active inference is a good lens through which we can approach this this question not only in relation to the modeling perspective but also the phenomenological perspective because something that I I didn't mention at least in the presentation but that's in the paper is that these anticipations these tcid anticipations can be understood at not only an effective level but also at an epistemic level so for instance Fel would say that uh sometimes you're walking down the street and see what might be the shape of a human being a person and you have a set of anticipations related to that you may I would say you anticipate uh you know talking to the person or you know

engaging different uh activities that are related to human beings but as you come closer there's something odd about this shape it might be either a human being or a doll K suggests and the important thing here is that this this potential this patients kind of are modalized they become doubtful whoo would say and that's very reminiscent to me to what happens in in active inference so when you know the agent is updating its inferences it's all it's a matter of trying to not only take into account temporal infolding of the world but trying to get a hold of it right it's trying to recognize what is going on outside so there's the constant of change because the world is constantly changing and it's also clear in phenomenology and not only the world is constantly changing but the subject Consciousness is always changing this constant flow that's what defines Consciousness from a phological perspective but what I like here is that what links these two domains together at the phenomenological level is affection very literally this embod Dynamics related to the the world and active inference does something very similar you know this inference related to the policies are what keep the generative model and generative process together so there's this constant kind of correlation and and and Loops between the agent and the world or the environment that are also find found in phenomenology so I would say that if you ask me about analogy and I think that my coauthor of the other paper we discussed here last year Zach he would he would gladly say that this is a process on and I I agree with that I agree with that it's the constant is change so one last Quick followup question if if it is about effect then I I know that I can be conscious I also know that there are definitions of Consciousness but I'm not sure that the definitions of consci Consciousness carry an effect I think it's only while we're aware of being conscious that that effect condition seems to apply would you agree what do you mean by an effect and in in the in the classic sense of it's it's here are my emotions these are the effects that I'm sensing on a on a moment by moment basis so this is again goes back to this so we seem to have this really interesting capacity to be able to externalize things and strip out effect because I know poetry can make me have all of these great Sensations but once I put the the poem down and go away they don't follow me right they stay within the the symbols on the page I I carry I carry the sense of Consciousness around with me that's the being part and so this is again I don't have an explanation for this but I just have a lot of Wonders around it yeah okay so that kind of comes together with the other thing I didn't mention too much in the presentation which was the whole thing about General feelings so affectivity as I understand that through the the work of javanna Coletti is this kind of General uh it's this lack of indifference towards the world and this can be understood in very very different levels it's it's a hierarchy if you want to put it like

that so emotions I would say are quite complex you know you have to have an intentional object most of the time you have to have certain bodily Dynamics going on there's an appraisal there's a lot of stuff going on it's it's this fascinating combination between cognitive and bodily components right moods are also super super complex but I would say even at the most basic level this this interplay between affection in the *ceran* sense and temporality what you get is not only you know because you're always anticipating Bly anticipating the world sometimes your anticipations might be completely doubtful right maybe something happened and this kind of change the whole anticipatory Dynamics and it's not only an epistemic thing it's not only like maybe I should doubt about everything there's also a lack of trust and again like Psychopathology shows this quite clearly I mentioned depression but there's also a phenomenological research on trauma and what happens in trauma people who suffer U uh severe severe trauma is that they cannot they can no longer trust other people and the world and it's not that they explicitly think about that fact it's simply that the world appears to be dangerous or you know as something that might attack me and this is an effective domain that it's always kind of I wouldn't say coloring but it's always kind of constituting or it's always uh defining or shaping how we experience the world so this kind of very general basic basic domain might change in relation to emotions which are higher and moods but it's always that we're always effectively coupl to the world and this if if you buy into my whole picture this effective domain is also active in a very literal sense okay I have a slightly different question how do we go into that micr temporal experience like how do we gird or prepare ourselves or do we go in holding our hand or are we holding a wall or how do we enter into perhaps the the direct investigation of some of these claims that are made about in in a way how our experience arises from our bodies I think that the best way to do that is trying to keep a very tight relationship between lived experience with phenomenological analysis and n only the classical ones we can also think about qualitative phenomenological uh research on lived experience made by people like um andan loots and and others and we can but we can also have to we also have to hold together the empirical domain and that's the whole point of neurop phenomenology which I truly buy into so on the one hand a phenomenological analysis like the one I just proposed may actually guide how we understand neurodynamics because you know if you look at say an fmri you see all these neural Dynamics going on but by themselves they don't tell you anything at all right there're just caal processes you need to correlate this to something going on at the at the at the experiential level it's not only the fact that we have data but we have a very very rigorous very strict uh description of what's going

on here and this may tell us something about how to interpret and how we understand neurodynamics but these neuro Dynamics will also probably give us some light on the experiential Dynamics uh because maybe there's something that doesn't fit too well with phenomenological distinctions and then we probably have to come back and reanalyze experience and that's I think it's more of a kind of social process right it's we're talking about a social communities not there's this constant idea that well not constant but a lot of people reject these phenomenological approaches because they think it's just introspection as in you know early 20th century psychology and it's not really that we have a very strict methodology and not only that we have like quote unquote peer review there's a lot of discussions and I'm entering in a specific discussion here saying you know you talk about temporality and effectivity as distinct no no no they come together but this is ongoing and I think I truly believe that there's no end to this which is kind of a beautiful thing as well but it's it's trying to maintain this circulation between empirical and phenomenological levels if we don't keep them together we're missing part of the point another very interesting thing in the hero on time Consciousness slide you had the kind of trace of the past sedimentation and then the cone of of protention and then beneath that it said direction of the flow and the flow was going to the left pointing back whereas a common time visualization would be like well time is going forward so what is pointing backwards and why don't you need a forwards pointing Arrow I wonder whether I should have just do like a two-way Arrow because a friend of mine read a draft of this paper before I published and and she asked me is it the other way around and I understand you all mean because that's true that's kind of and I'm very like forward looking that's part of the point what we talking about you know intentionality future oriented but I think that when you analyze it in the way us does it what you not notice is that the stream of Consciousness is sinking into the past is like a spear that is directed towards the future and that's intentionality but it's always like a river that is sinking into the past and that's the reason why uh the arrow goes to the left but you might as well put it in to the right and you have you know it's just Speer heading to towards the future Daniel's not pointing out that he knows that wayfinders have to get back before they can go out he's he's he's conveniently leaving that off he he recognizes why the arrows going the direction it is it just in that representation reminded me of just like putting down the track as we head forward like building a cliff that's always being peaked at over but as we proceed it's like the ledge just keeps on extending as we move forward and that creates the solidity of our retention um I think so there's also this metaphor of sedimentation right that some phenomenologists use quite a lot they talk about

how the what we experience it's of course retained but time is sedimented it kind of metaphorically uh builds the foundations the soil of what we experience and this changes our whole experience so we may understand habits in that way we experience some some events over and over and this set it sedimented onto our Consciousness that's a metaphor of course and that kind of generates habitual ways of engaging with the world and I think that there's something valuable about emphasizing of this this sinking into the bit without downplaying the fact that intentionality is directed future sorry I I'm gonna I I really apologize I really wanted to participate in this one one Diego because I I think your work is is fabulous but I now have to run and go and deal with some politics in the wilderness so I'm going to check out right now but thank you thank you again for uh your fantastic work thank you as well thank you for being here at least for a bit bye-bye okay another kind of question so you distinguished the active object intentionality from the operative passit or implicit intentionality so how do we make sense of a passive intentionality is it like a desire that we don't have a control over whereas an active desire was one that we kind of selected Upstream to want to want or or what is the difference between the active and the passive intentionality so that's the husan terminology I don't think it's very popular nowadays so phenomenologies often talk about the reflective and pre-reflective levels so the reflective is like thematizing explicitly something whereas pre-reflective is not like in a temporal way it's more like a condition for reflection but what I like about the distinction between Pacific activity is that it also shows that act like act intentionality or object intentionality requires uh degree of activity from the subject or from the agent and you know for you to attempt something you are moving your head you're moving your eyes you're you know moving your body even if you do that somewhat uh by m reflex still a form of activity right there's some something that you as an agent as an embodied agent are doing but regardless of that there's always this openness to the world we cannot stop that unless we die and we don't want you do die to die most of the time right it's passive in that sense it's it's kind of almost instinctual and I wonder whether we may talk about instincts at that level and actually there's this very talking about uh habits again there at some point H talks about two levels of passivity and I don't know whether he kind of goes on with this but I like that because he says that you know there's like the primary passivity which is this very basic effective Dynamics but there's like a secondary passivity which is pris the habits it's like a second nature to us and it becomes passive because you know when you're very habituated to something you don't think about it you don't have to do any kind of reflective process it just kind of happens and that tells a lot about how we engage with the world at a

phenomenological level for sure and it kind of suggests like the sedimentation of what is potentially an active intention in a moment like if we were practicing archery what might be active in a moment or what somebody might cue us to think about can become sedimented and that's kind of the process of our actions becoming habitual or our behavior in the moment over time it's like you're flowing this color of sand over the river eventually it sediments that color without needing further flow of that color exactly and that's a very instinctual perhaps and very implicit way of understanding learning right because that's what happens when we're learning about something at the beginning especially skills think about bodily skills if you want to learn to play the piano that is very difficult you have to very explicitly move your fingers and it's awkward and it sounds horrible but over time these movements become habituated and we may talk about bodily memory or something like that but it's the same idea you know it becomes second nature to you it's it becomes almost passive. H yeah and thinking about learning a language or math or something it's almost like an applied area of This research agenda would be to discover potentially to validate some differences in learning exercises and then maybe to suggest new kinds of learning exercises based upon what could be attended to in a moment that if it were sedimented could then be later used as an implicit jumping off point like an example coming to mind is let's just say like an integral let's just say that the rule for the integral is like there's always a number above and below but before even going into what those are you just sediment the understanding that there's always a number above and below it and kind of swallow that and sediment it and accept it and then show what those numbers mean having accepted that they must be there rather than try to show a new type of symbol and suggest that the symbol is a certain way because the math has to be a certain way and try to co-sediment um a visual understanding with a sort of analytical understanding but maybe some parts could be sedimented in a different order yeah I wonder about that last bit I mean I don't know nearly enough about learning more gener but I think it sounds very roughly correct to me because if you think about maths for instance you know when you have to solve an equation and you already are skillful at maths you still have to think about you know arithmetical calculus processes right there's an active explicit level but this is always grounded in a uh and understanding with which is already kind of second nature to you you already know as you said in an integral there's the number up top the one at the bottom is a given and you kind of already know what's the function of it all it doesn't give you immediately the solution to the equation necessarily there's an active process but it's always built on top of a background of Habitual uh processes

which is precisely what a per of intentionality is trying to capture and and I mean I I I haven't thought too much about the relation between maths and this but what I like about as as an embodied person as a person who likes embodied Cog cognitive science uh what I like about operative interality is that it's clearly deeply embodied and there might be a way of connecting you know these very abstract theoretical processes like doing math to bily processes and in active inference that would be literally the active fit of active influence isn't yeah potentially there's a that makes me think of a Memory Palace and leveraging spatial analogies and protentions I could walk out of this room into another room and I'm going to put a fact there or put a memory there um or associating a smell or a sentiment with another area of information and then sedimenting that not so that it is directly recallable but I guess just when I when I again to the the trace and the protention funnel I think about Mike Levan and the time cone or the time diamond and we kind of in that visualization it's almost like the cone goes off in both directions so we have the perimeter in the moment which is like the extent in the now and then the past is seen as um I I think one way is probably like uncertainties over the past whereas in the uh who Roll On Time Consciousness slide it was like there's just a single trace so we could think of it though as uncertainties or more and more scope of causal precedence leading in like you know our grandparents and our great great grandparents and all this coming in through the um needle of the moments um and then like decision making is jumping off into the protention only to be answered in the sensory consequences at the next update exactly I that's what I think that it's a very nice connection between the active inference conception of of you know cognition and decision making and these more phenomenological uh description because what you get is precisely this kind of isomorphic uh mirroring almost between the structure of the updating of a generative model in active inference and the kind of Dynamics the temporal Dynamics you have in phenomenology and and what is kind of nice that as you make more and more complex your your gentic models you can introduce things like you know uh effective inference like Casper's work and I think that would be an interesting way of trying to capture some of the effective domain the effective dynamics that occur at the Phenom phenomenological level and that's precisely what I mean that you have to kind of make this constant circulation between different domains like on the one hand the phenomenology might be able to inform us how to um kind of organize and kind of uh think about our generative models and these generative models may actually help us understand neural Dynamics and that's kind of the whole point of what now we call computational neurophenomenology but as as a philosopher and a self-described phenomenologist I would

say you know we have to start by the phenomenology it doesn't necessarily mean that phenomenology trumps everything it simply means that it's our starting point because that's literally what we have that's all we have as experiential beings lived experience but we have to be very careful with how we analyze it and again it's kind of fascinating for me these similarities between the mathematical structure of active inference and these more formal Dynamics in phology hm it also makes me think about first learning about xenobiology and theoretical biology thought well biology is living real things so if we open the door to life forms that we haven't observed or hypothetical life forms what isn't on the table that's going to be too broad because we have enough to investigate with the forms that are here so then we have the starting with what we have and building out approach whereas some of this kind of like I don't know if it would be like Xeno phenomenology like thinking about radically different minds and systems which in some ways is an unconstrained Quest because it's going to be hard to get unique explanations and predictions about like the experience of an extraterrestrial that we haven't seen yet in that social Quest there are useful factors identified and in some ways it ends up being the people's different views on the plausibility or the relevance of a given speculation that reground even what initially could seem to be like a totally ungrounded question like what is it like to be a bat or what is it like to be a cloud or something you know I I I find that kind of interesting because that's part of what I've been what I did during my PhD it's precisely thinking how these phenomenological approaches and in active approaches plus the F might actually let uh allow us to understand better biological cognition broadly understood so I'm not I don't know much about kind of life as it could be but Life as We Know it it's clearly you know bacteria or plants or animals moving around and trying to survive right that's the basis of it all but what we what I what I would say is that for us to understand that we have to kind of assume that there's a degree of I mean very very broadly understood subjectivity in there I don't want to say that the bacterium is conscious or aware or something like that I don't think so but there is a sensitivity to the environment you know when the bacterium is swimming you know in chemotaxis swimming towards the glucose there is a sense in which the bacterian is sensitive to the environment is affected by it that's a very natural way of putting it and what is fascinating again is this kind of similarity between these very complex phenomenological analysis of human experiences and when we find when we can see kind of from a third or second person perspective on other living beings there's very close similarities and if we take very very seriously theeran claimed that the whole point of phenomenology is to disclose this invariant structures of Consciousness or

subjectivity or whatever you want to call it then there there's a case to be made that uh these invariant structures may actually speak of the at least the conceptual basis of what we may call biological cognition and I mean that's a massive project but I think there's something about that and I think again that active inference is it's an interesting kind of bridge between all these different perspectives because if you say that active inference is capable of say modeling you know biological Dynamics on the one hand and also uh experiential Dynamics as you know computational phenomenology claims then it's it's this kind of interesting formal bridge between say biology phenomenology and neurobiology Etc it's of uh it's it's a bridge it's a very helpful bridge I would say yeah the second person perspective it makes me think about the um biography of Barbara McClintock who's a biologist studied transposons and mobile genetic elements and the biography written by Evelyn Fox Keller is called a feeling for the organism kind of alluding to the career-long quest to have a feeling for corn genetics in this case and other genetic systems and so whereas the what is it like to be a bat is kind of like a third person firsters dialectic I'm over here third party to the bat and I want to get into that driver's seat and be first person in the bat and then what would it be like and so that's kind of like the first third first third I'm my first and I want to be its first so I'm its third but then the second person which I see in the second person Neuroscience area is like what is the feeling for and the relational component that is actually a kind of alternative to swapping first person perspectives because if it was just swapping first person perspectives you don't and you just go from I am person I am a bat yep versus you are a bat is engaging in the second person and that brings in the social component of the phenomenological project and distinguishes it in a way from a kind of solipsistic firsters eye agenda yes yes for sure sure for sure and I think that kind of project is quite radical in the sense that it has very interesting consequences uh in relation to how we think about science right because in a very kind of traditional I would say even dogmatic way we think about science as this strive for objectivity like the third person perspective as such and maybe that makes sense in in certain Natural Sciences maybe physics perhaps I I don't know but in something like biology you're dealing with another living being right and it's not in certain cases might be a third person perspective if you put it but you know when the biologist is looking at the bat or the bacterian or the amoeba or whatever there's there's a quote unquote social process going on you know I I the scientist I'm recognizing you the the biological system as a biological system I'm actually noticing something in there I may not know what it's like to be B but I know what it's like to be a living being we have a common ground and that's that at least from the human perspective is an experiential common

ground and that's what phenomenology breaks into the table a way of trying to analyze this but again it's a very radical change I would say because at least biology becomes kind of this quote unquote social science not in the sense of Sociology or anthropology but it's a matter of I'm actually engaging with this this biological system and if we build from there at least at the level of analysis for organisms maybe not genetics and you know or population but at least at the level of organismic existence I think there might be some ways of reconceptualizing our understanding of life Wow like on the overemphasizing first person we have the cartesian thought experiment hyperbolically overemphasizing the third person denying interiority we have behaviorism only standing outside of the room and never asking internally and there's another bridging path that is very provocative that biology would be a social science is because yeah that will be interpreted in a disciplinary setting but social being like composite or Collective yeah and co-active which is exactly where it feels like biology should be yeah exactly yeah I say social science in the broadest of Senses not I wouldn't say that again there's a very strong connection in sociology or anthropology and biology but social in the sense that again there is a social connection between the the the scientist and its object we're talking about two quote unquote subjects relating to one another they're Co co- enacting a world in a sense and that's that's a very inactive conception like if you you look at the works of Francisco or Evan Thompson EO their ideas that are very consistent with this approach and I think that phenomenology may also bring interesting uh tools to understand this because you know there's a whole literature and phenomenology about sociality inter subjectivity and it's usually emphasizing how we understand one another as human beings it's not Sol system but the question is what happens if we try to apply this analysis to our understanding of other living beings not only human beings and that opens up a whole new you know domain for research I think yeah and perhaps even speaks to the unhealed Gap with like ecology is everything that happens on earth except for human activities that's why it's human ecology or biology is all living systems except for the social because those are the human and so there's kind of this tension with kind of like are humans natural and are we part of the fields and methods that should be approaching any other kind of mammal Andor do we rely on more transcendental methods that wouldn't be empirically useful in studying like an insect at first pass sure I I guess the question would be if we take this kind of transcendental path if you want to call it that way how can we let's say operationalize this into a more empirical project because as a philosopher I'm happy to say we should do transcendental phenomenology and make S about transcendental biology but I'm also quite aware because I work in interdisciplinary context that

that's not enough that gives you a very interesting and I would say very strong theoretical basis but that's not empirical science right how can you step from these theoretical claims to making you know testable predictions and hypothesis that's the question and that's a fascinating question I think yeah and that well maybe you've outlined an alternative meeting point which is rather than finding a common molecular basis between humans and fruit flies I mean we know that we have proteins carbohydrates we know what reductionism is going to find when we're turned into dust but you're in a way outlining a path of saying well certainly something affects a fruit fly and it doeses have a protention might be a shorter or different shaped protention but by rather than trying to materialize The Human Experience only and find a common grounding on the materialized by having a common grounding in the time Consciousness then even the bacterium can be jux opposed with the human and not on a material comparison not just because we have shared material or DNA but because we have a shared living autopoietic embedding yeah exactly there's there's this continuity between what it's like Quon to be a bacterian and what it's like to be a human it's certainly not identical we cannot reduce our kind of experience to whatever kind of interiority uh it's there in the in in a bacterian we cannot do the other way around either we can we we recognize experientially these Dynamics right and I think there's actually empirical evidence for this kind of processes we I've read about you know quote unquote cellular memory how bacterian when they're swimming in you know a a milu environment there seems to be a very very minimal uh degree of memory or historicity and that explains a lot of behavior we may have even um mechanistic approach to this we can explain this quote unquote minimal memory from mechanistic perspective and I'm happy to do that because that's empir bit but what is interesting that we recognize that because we experience that in a perhaps more complex way but it's the roots of what we call temporality and affection affectivity it's at the roots of life and again that's that's out is that's Matana and I truly believe that if you buy into the phenomenological conception you will naturally tend to this more inactive conception all awesome well do you have any last kind of comments or thoughts uh no no no not really this this was been a very good uh discussion I like this thank you yes well we also um learned a lot from your work in our recent shared protentions paper and it's just exciting to see all these threads and and complimenting pieces the puzzle yeah there's there's an ongoing kind of new research field that is trying to link all the different concepts and I think that's very important because I'm kind of amazed at the fact that it might I think it has the power to change some of our ways in which we understand and do science uh hopefully we'll see to find out we'll find out I mean that's what

else can we do wouldn't expect any other answer wouldn't prefer any other answer um thank
you very much so good luck with your research and see you next time thank you thank you
Dan bye bye bye